

Introduction to Computer Vision

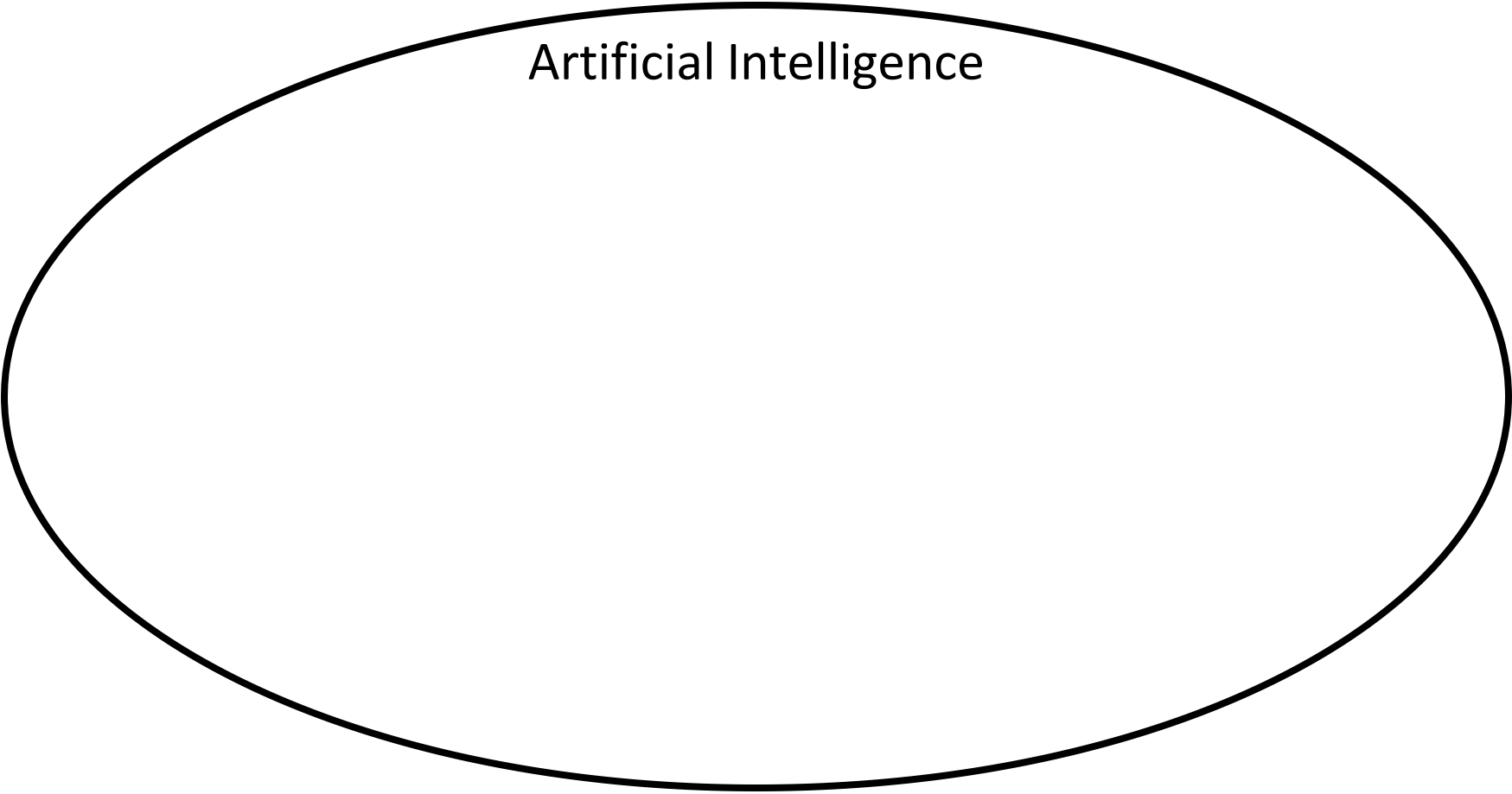
Course Code: CSC 4254 Course Title: Computer Vision & Pattern Recognition

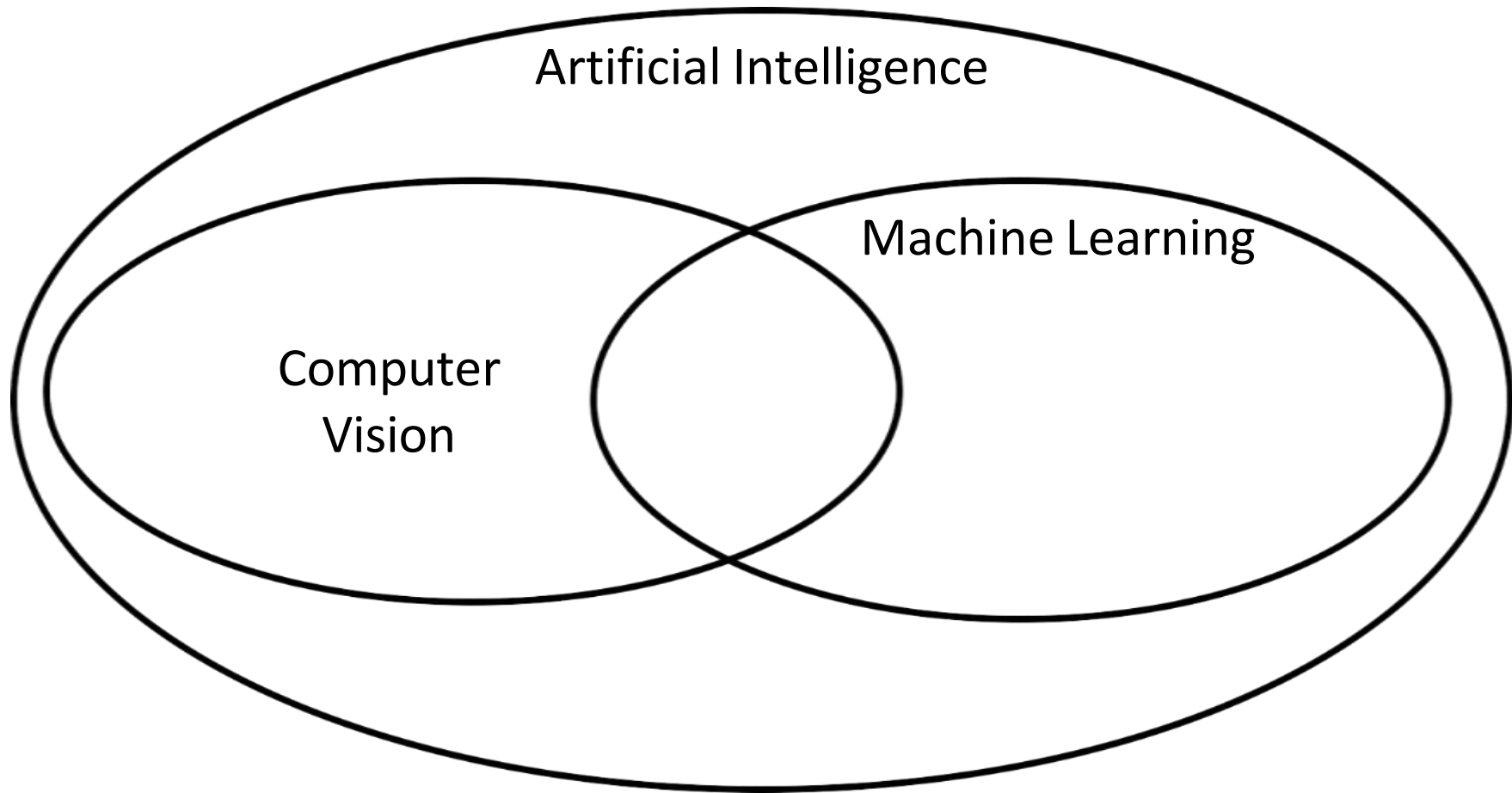


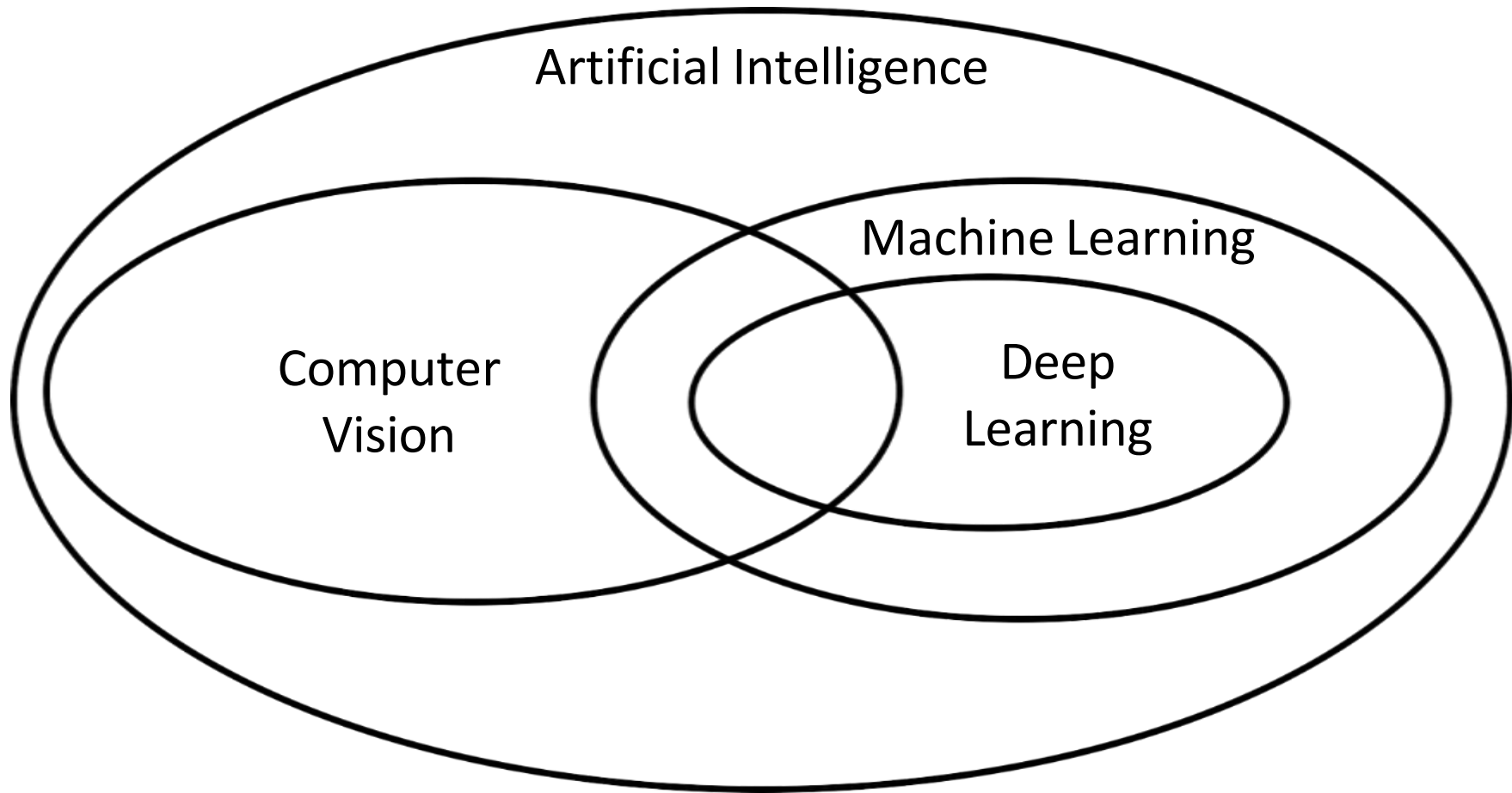
Dept. of Computer Science
Faculty of Science and Technology

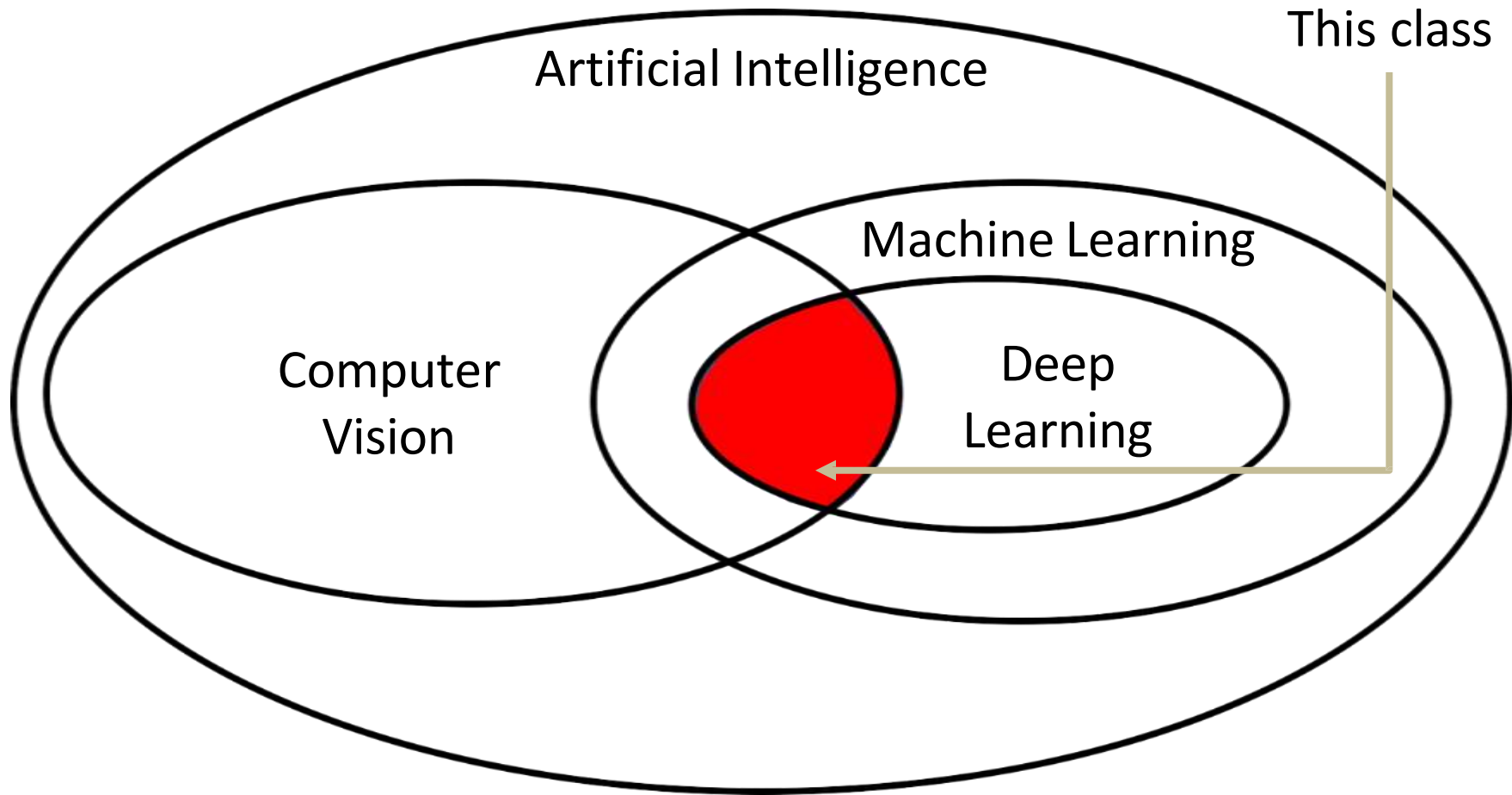
Lecture No:	1	Week No:	1	Semester:	
Lecturer:					

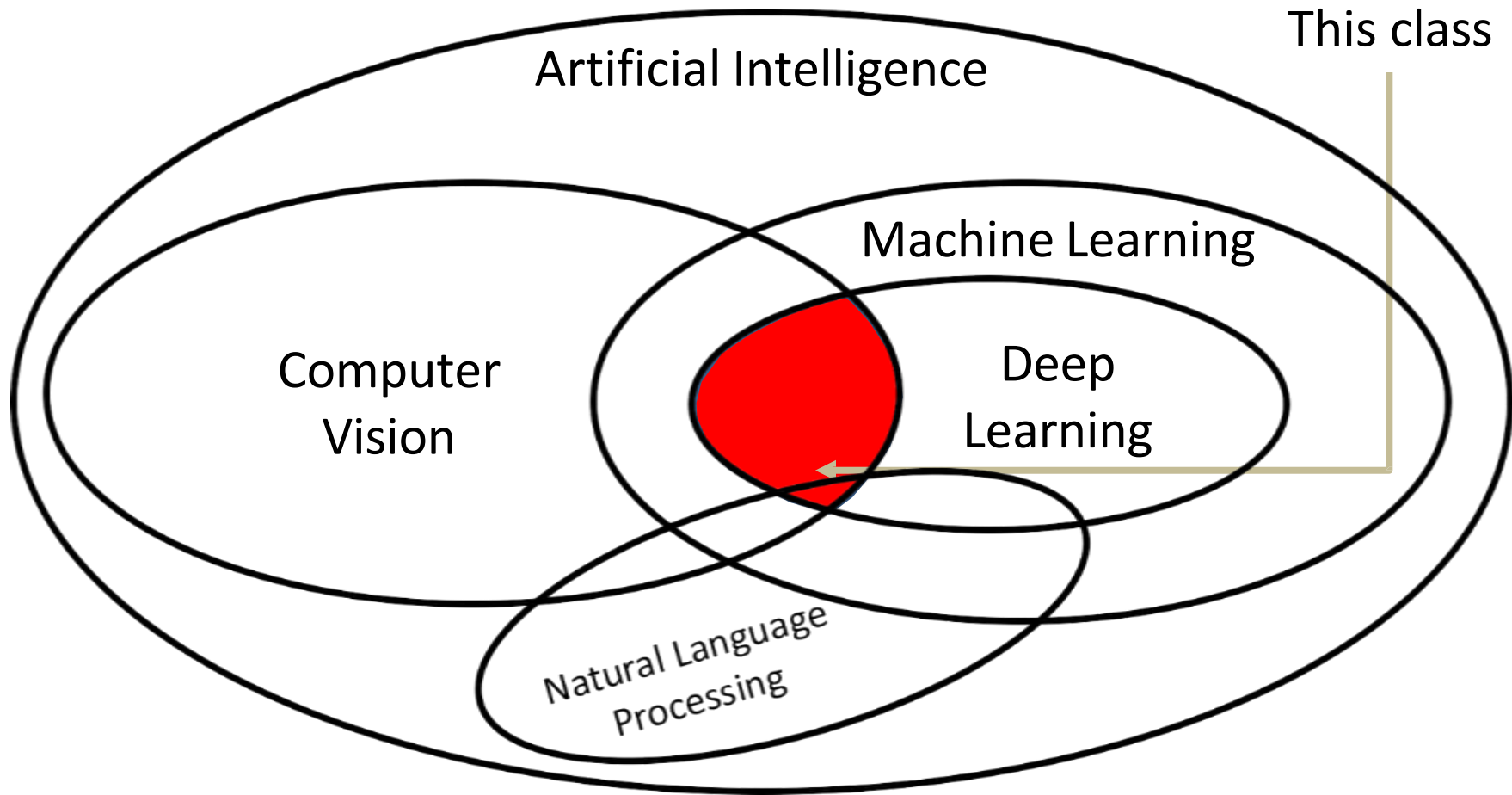
Artificial Intelligence











This class

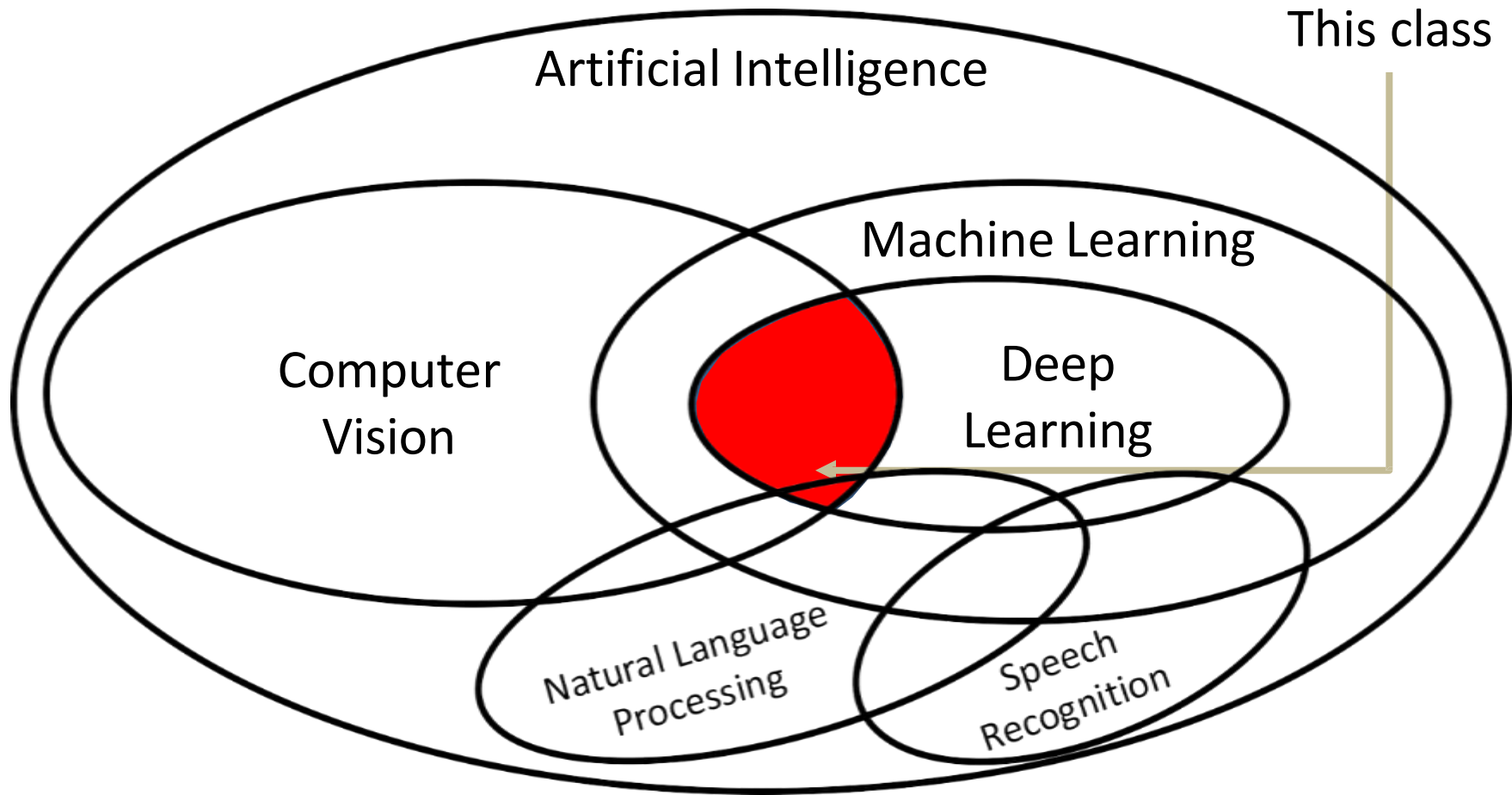
Artificial Intelligence

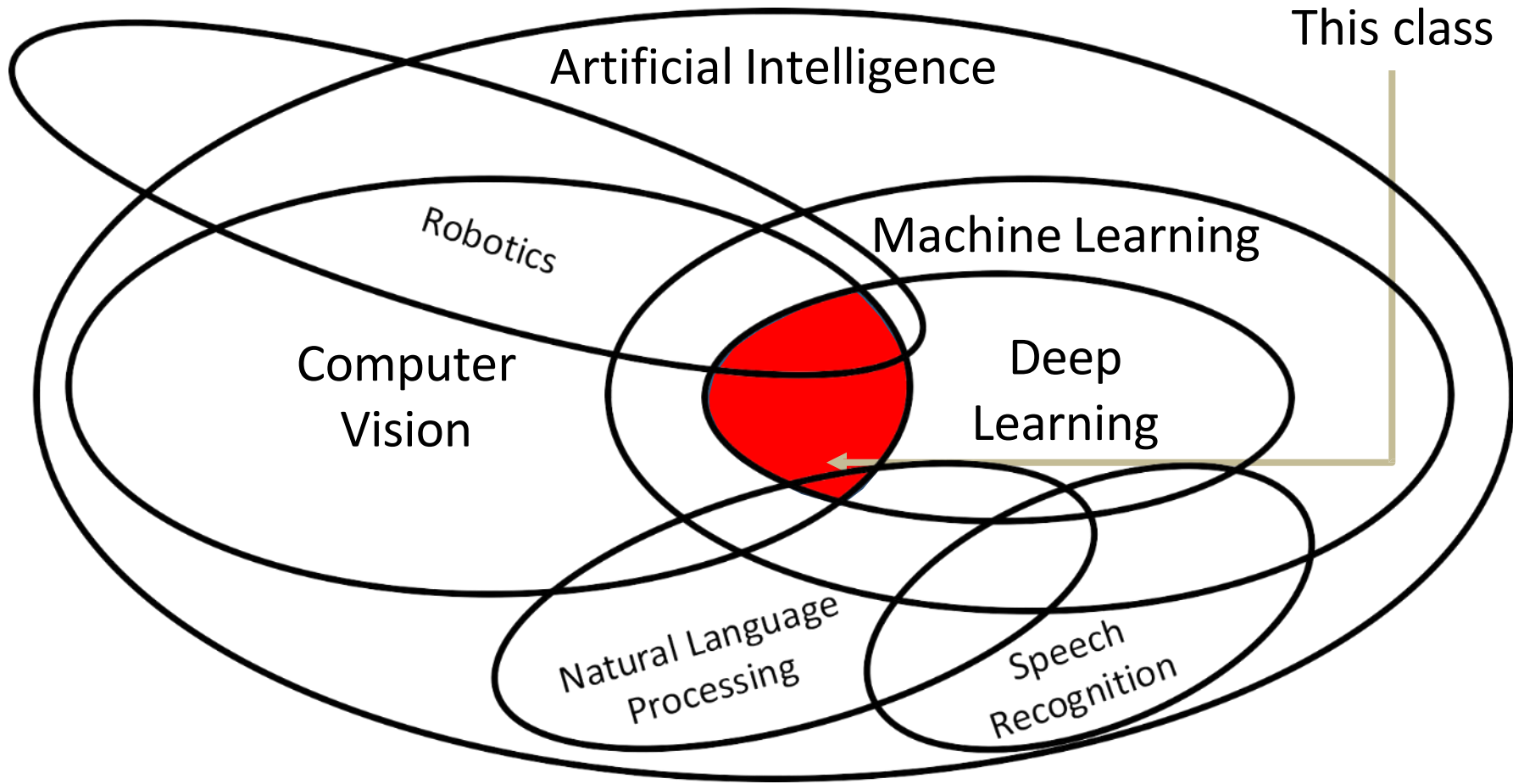
Machine Learning

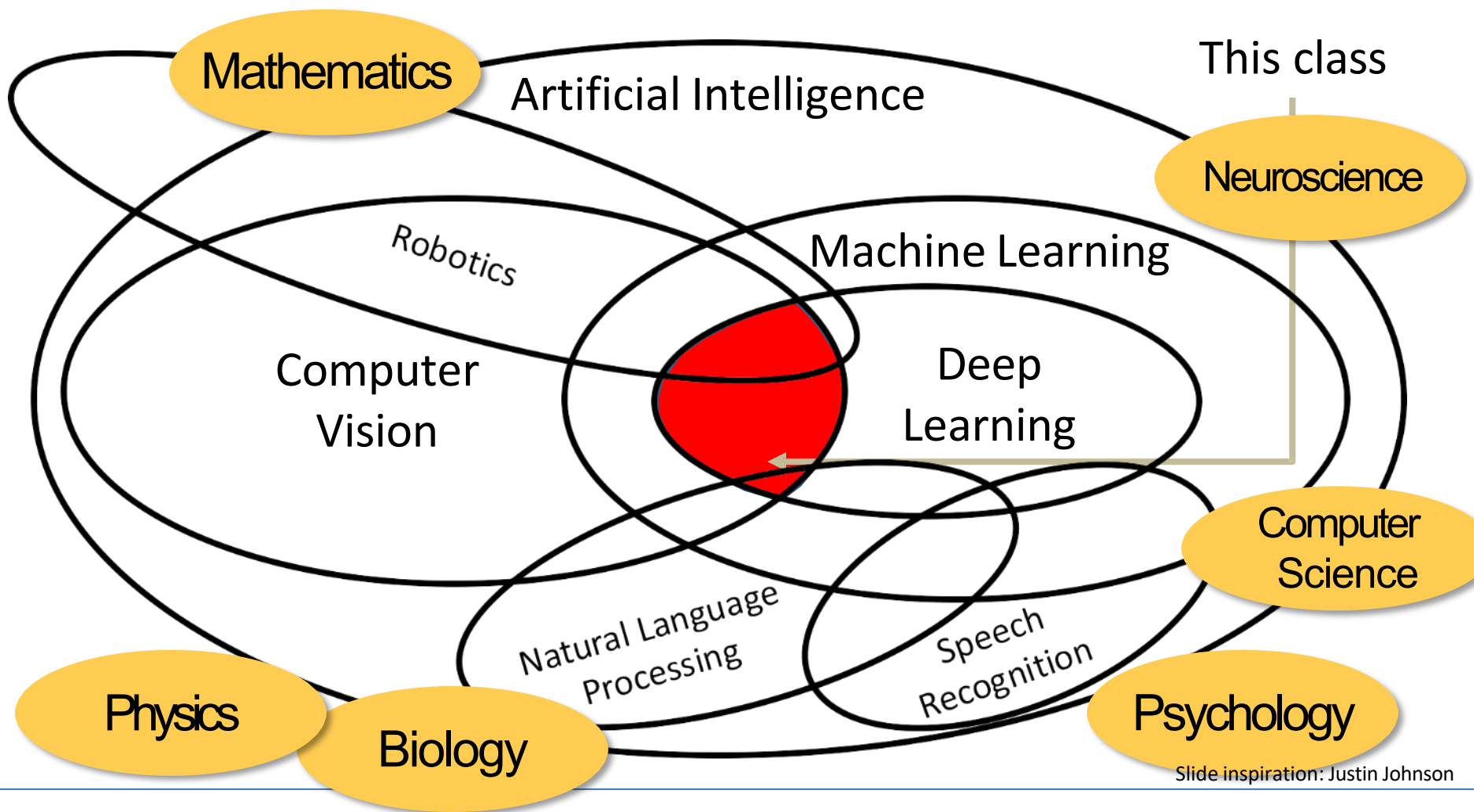
Computer
Vision

Deep
Learning

Natural Language
Processing







Slide inspiration: Justin Johnson

Today's agenda

- A brief history of computer vision and deep learning
- Course overview

Evolution's Big Bang: Cambrian Explosion, 530-540million years, B.C.



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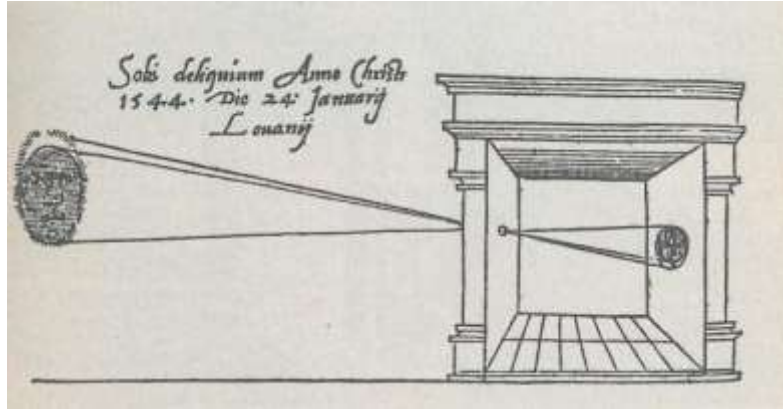


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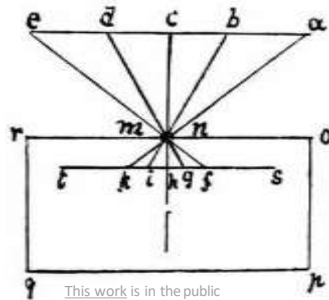


Camera Obscura

Gemma Frisius, 1545



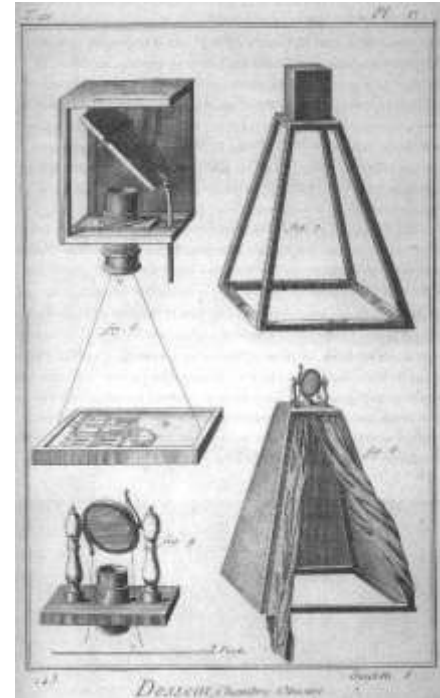
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Leonardo da Vinci,
16th Century AD

Encyclopedia, 18th Century



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Computer Vision is everywhere!

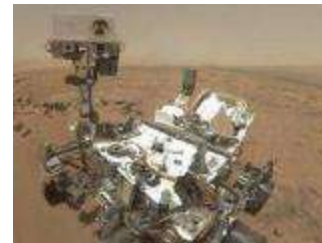


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Left to right:
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Bottom row, left to right
Image is CC0 1.0 public domain

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changes made

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Where did we come from?

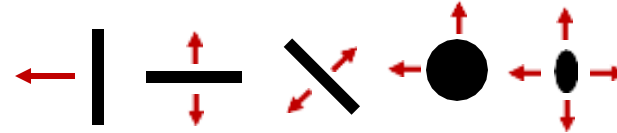
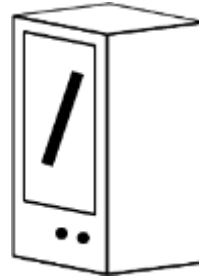
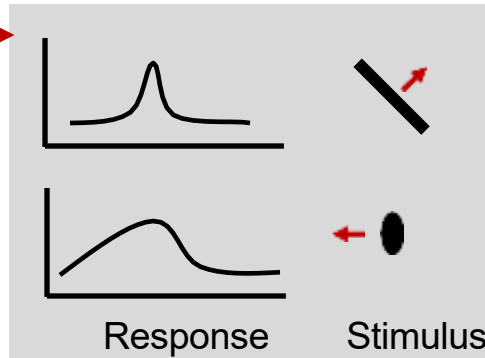
Hubel and Wiesel, 1959

Measure
brain activity



Cat image by CNX OpenStax is licensed under CC BY 4.0 changes made

1959
Hubel & Wiesel



Simple cells:
Response to specific
rotation and orientation

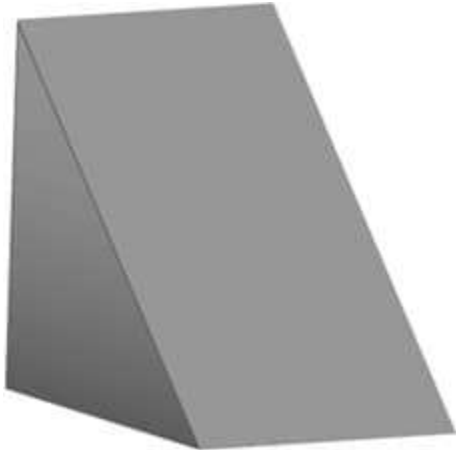
Complex cells:
Response to light
orientation and
movement, some
translation invariance



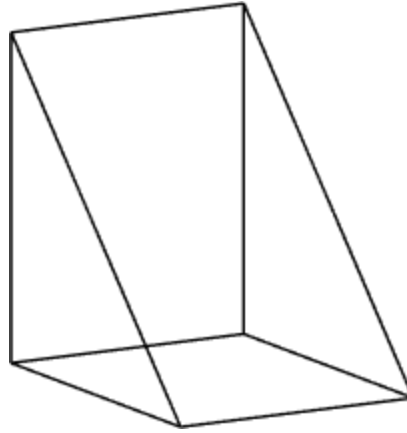
No response

Slide inspiration: Justin Johnson

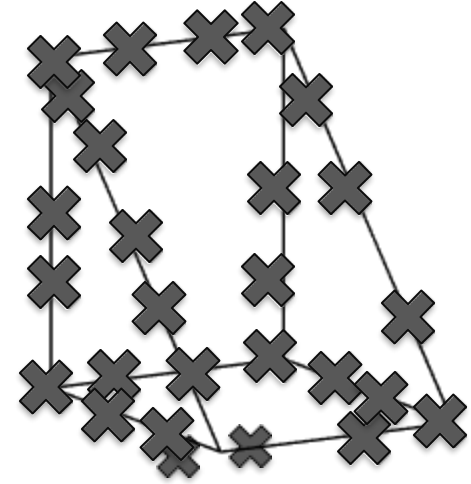
Larry Roberts, 1963



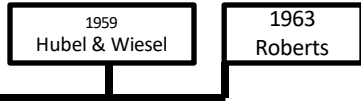
(a) Original picture



(b) Differentiated picture



(c) Feature points selected



MASSACHUSETTS INSTITUTE OF TECHNOLOGY
PROJECT MAC

Artificial Intelligence Group
Vision Memo. No. 100.

July 7, 1966

THE SUMMER VISION PROJECT

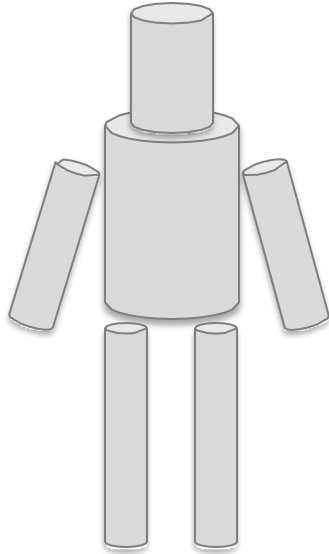
Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".

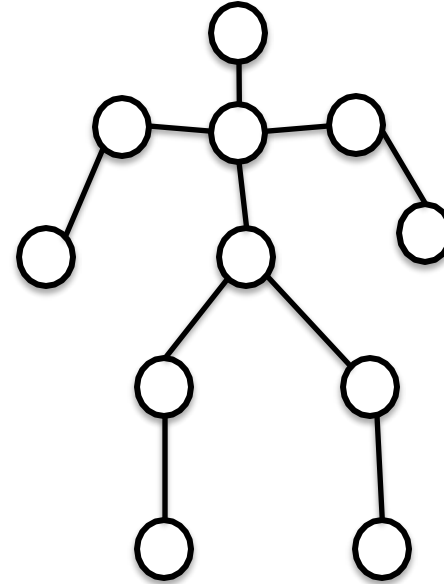
1959
Hubel & Wiesel

1963
Roberts

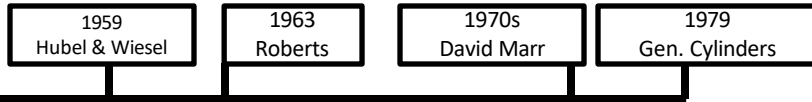
Recognition via Parts (1970s)



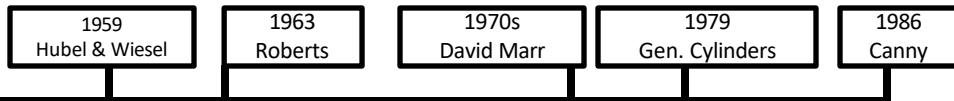
Generalized Cylinders,
Brooks and Binford,
1979



Pictorial Structures,
Fischler and Elshlager, 1973



Recognition via Edge Detection (1980s)

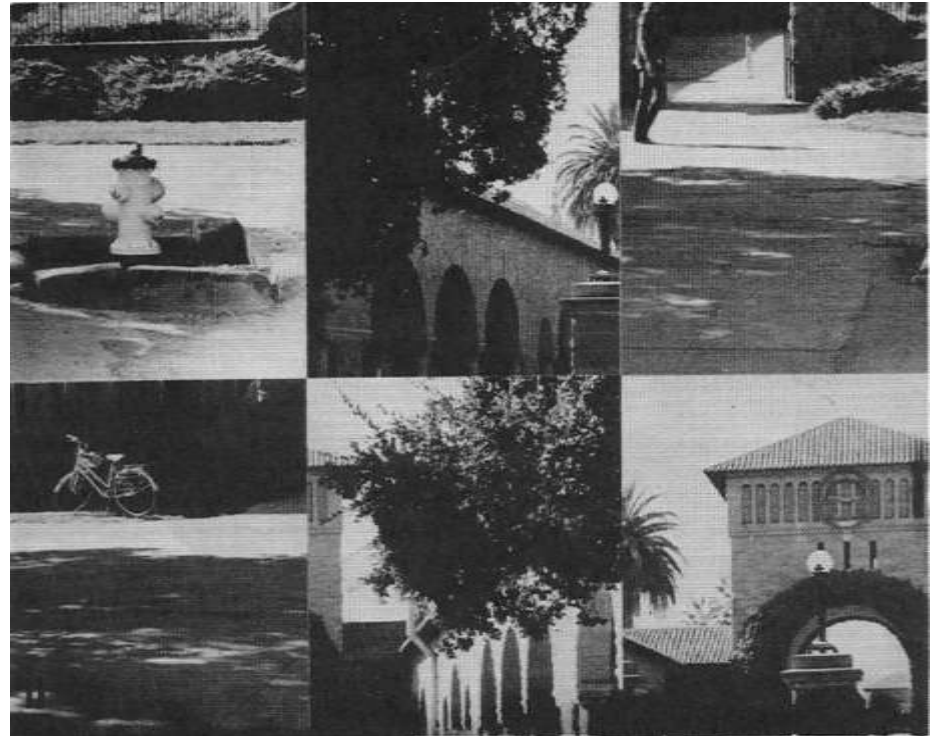


John Canny, 1986
David Lowe, 1987

In the meantime...seminal work in
cognitive and neuroscience

Perceiving Real-World Scenes

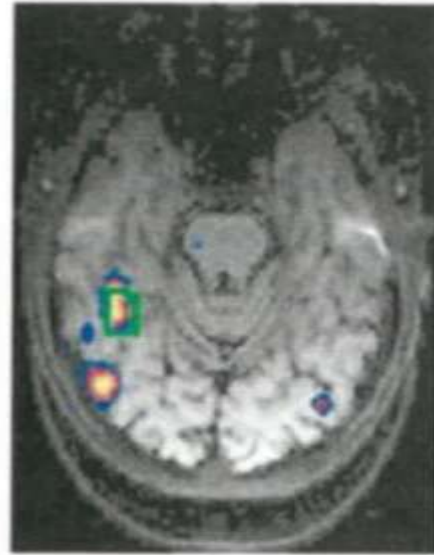
Irving Biederman



I. Biederman, *Science*, 1972

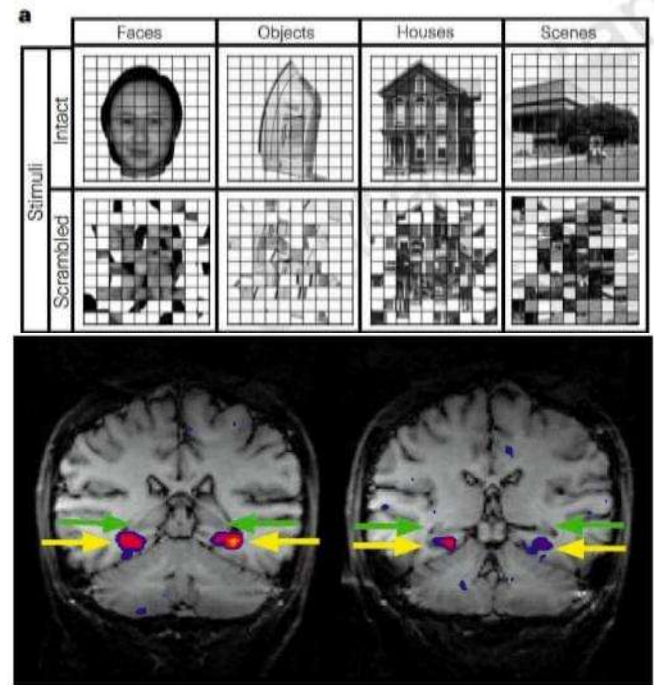
Neural correlates of object & scene recognition

Faces > Houses



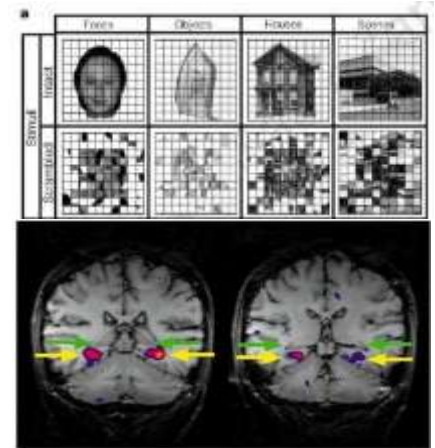
% signal change

Kanwisher et al. J. Neuro. 1997

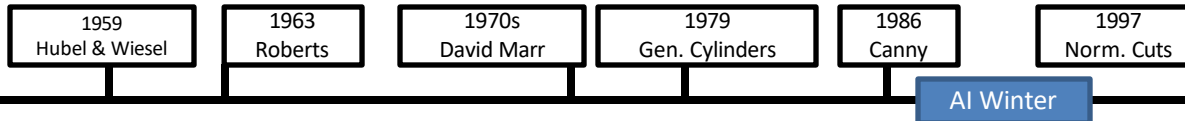


Epstein & Kanwisher, Nature, 1998

Visual recognition is a fundamental task for visual intelligence



Recognition via Grouping (1990s)



Normalized Cuts, Shi and Malik, 1997

Slide inspiration: Justin Johnson

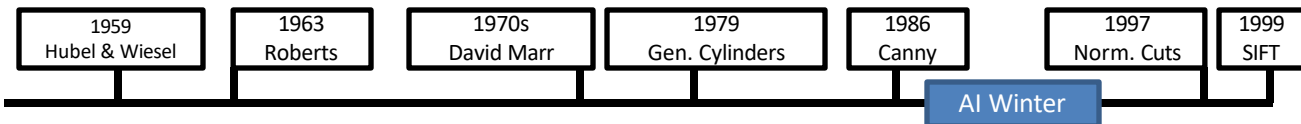
Recognition via Matching (2000s)



[Image](#) is public domain



[Image](#) is public domain



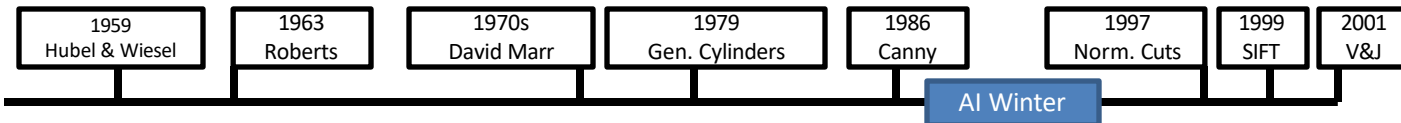
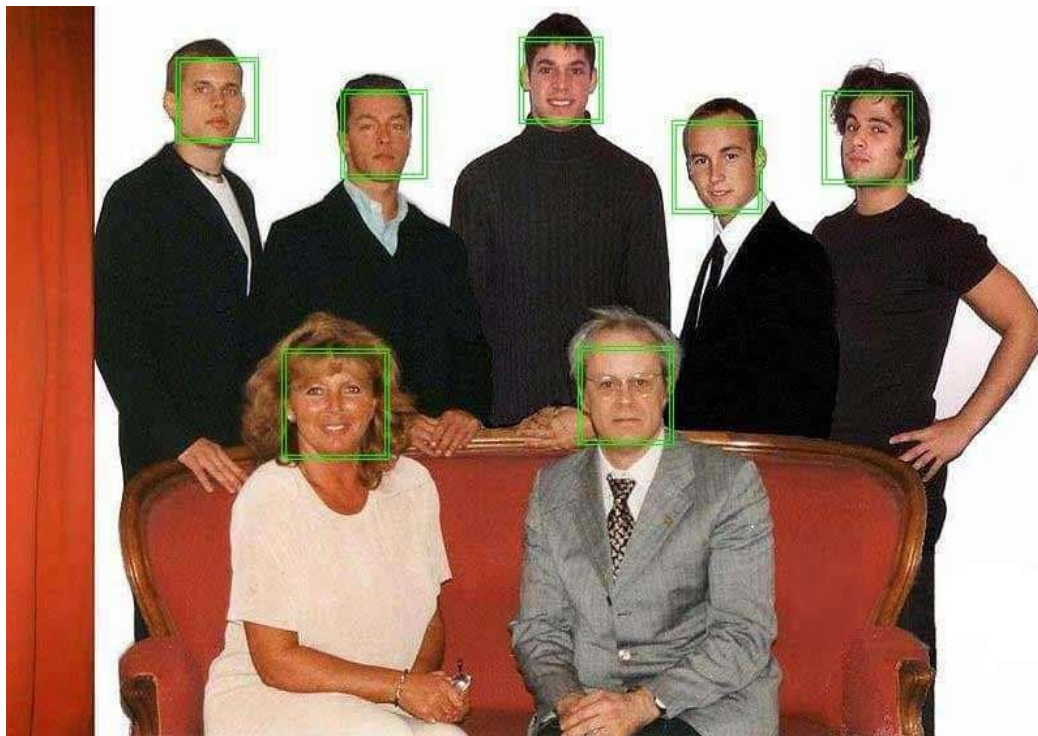
SIFT, David
Lowe, 1999

Slide inspiration: Justin Johnson

Face Detection

Viola and Jones, 2001

One of the first successful applications of machine learning to vision



Caltech 101 images



Pictures of objects belonging to 101 categories. About 40 to 800 images per category. Most categories have about 50 images. Collected in September 2003 by Fei-Fei Li, Marco Andreetto, and Marc'Aurelio Ranzato. The size of each image is roughly 300 x 200 pixels.

PASCAL Visual Object Challenge

Image is CC0 1.0 public domain

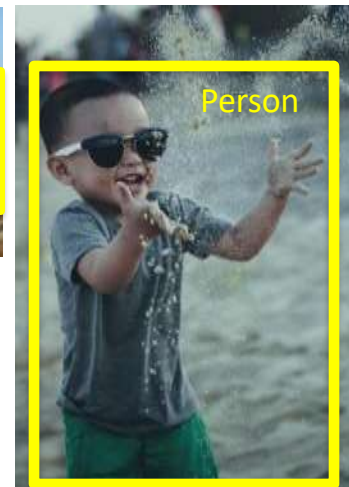
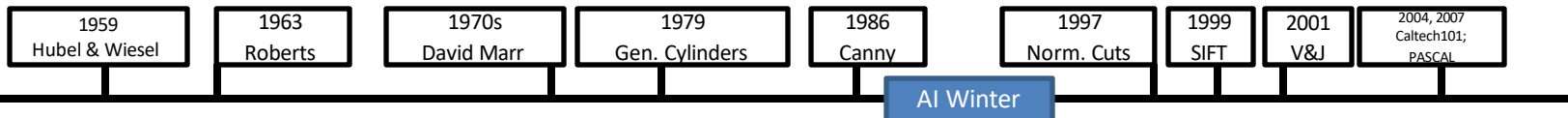


Image is CC0 1.0 public domain



Slide inspiration: Justin Johnson

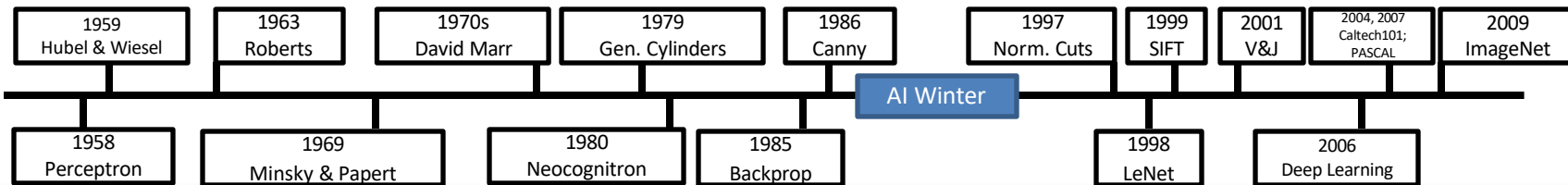
IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:
1,000 object classes
1,431,167 images

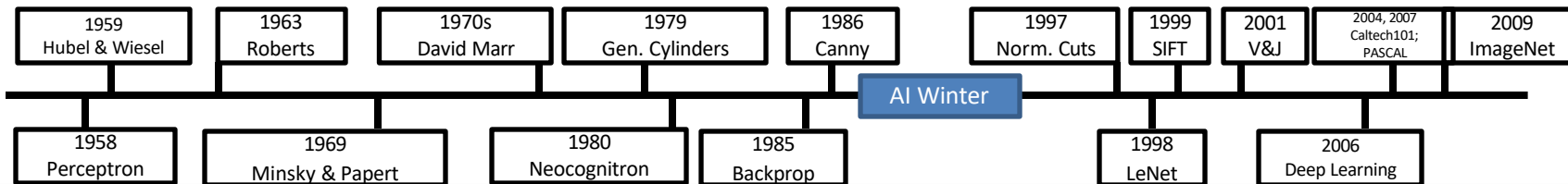


Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle

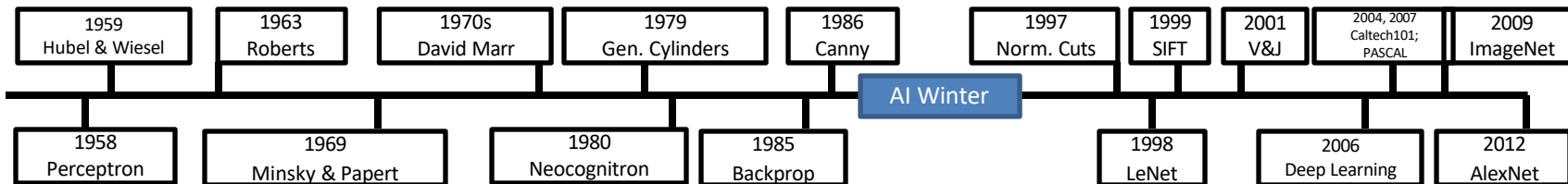
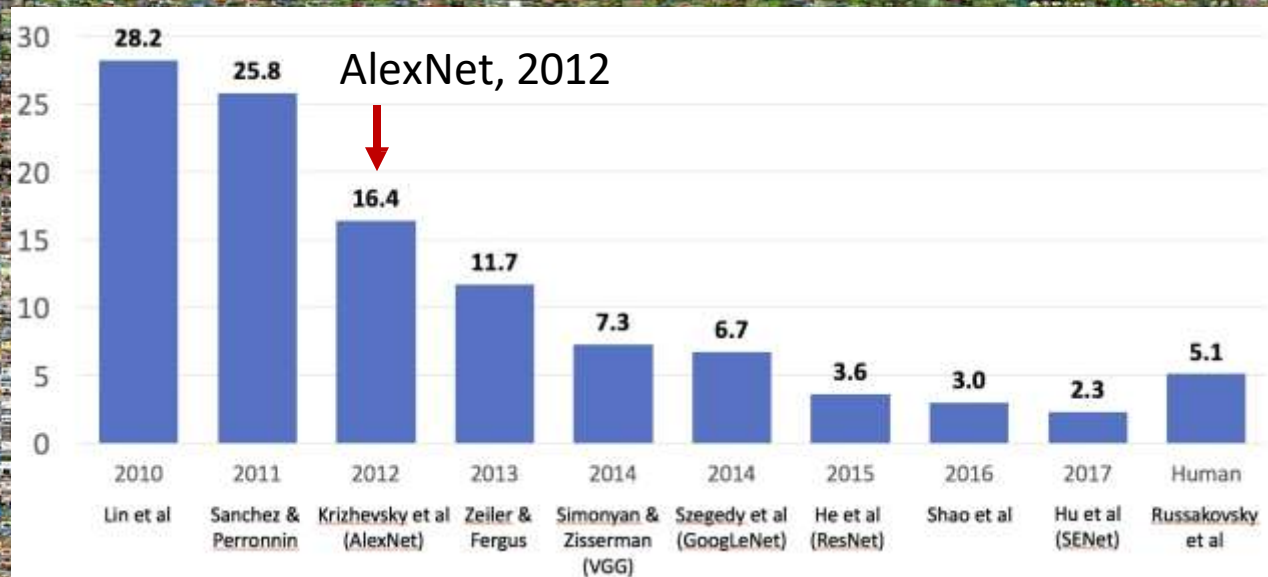
Deng et al, 2009
Russakovsky et al. IJCV 2015



IMAGENET Large Scale Visual Recognition Challenge



IMAGENET Large Scale Visual Recognition Challenge



2012 to Present: Deep Learning is Everywhere

Image Classification



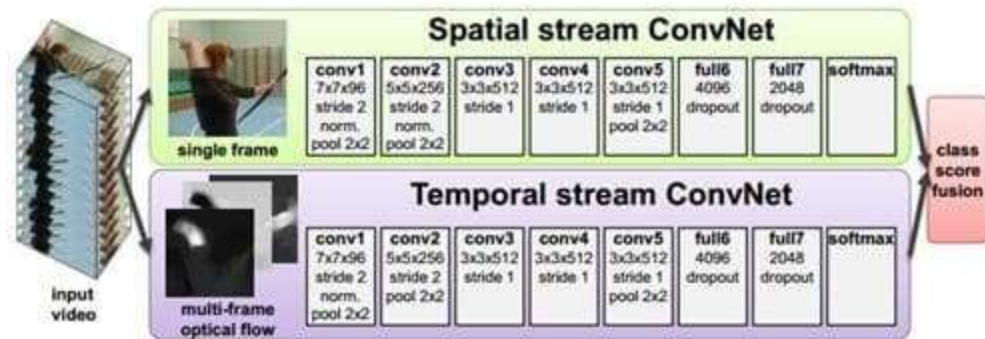
Image Retrieval



Figures copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

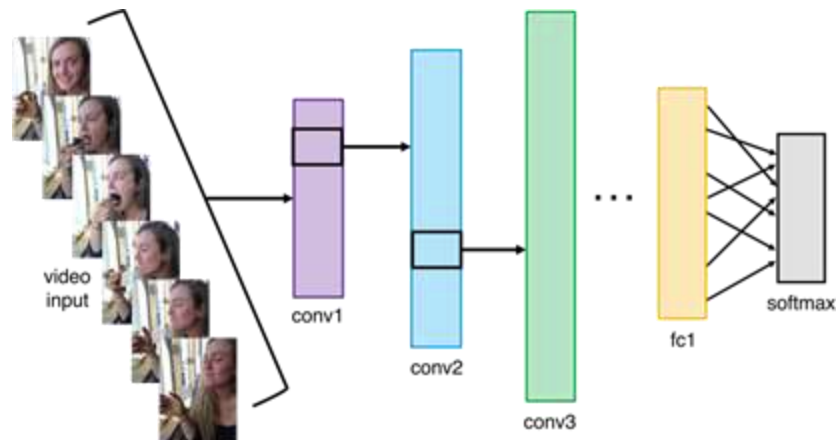
2012 to Present: Deep Learning is Everywhere

Video Classification



Simonyan et al, 2014

Activity Recognition



2012 to Present: Deep Learning is Everywhere

Pose Recognition (Toshev and Szegedy, 2014)

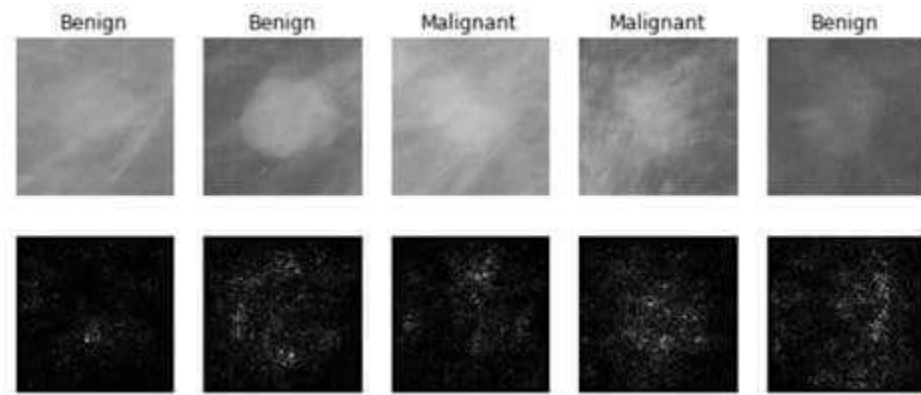


Playing Atari games (Guo et al, 2014)



2012 to Present: Deep Learning is Everywhere

Medical Imaging



Levy et al, 2016 Figure reproduced with permission

Galaxy Classification



Dieleman et al, 2014

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Whale recognition



[Kaggle Challenge](#)

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2012 to Present: Deep Learning is Everywhere



*A white teddy bear
sitting in the grass*



*A man in a baseball
uniform throwing a ball*



*A woman is holding
a cat in her hand*

Image Captioning

Vinyals et al, 2015

Karpathy and Fei-Fei,
2015



*A man riding a wave
on top of a surfboard*



*A cat sitting on a
suitcase on the floor*

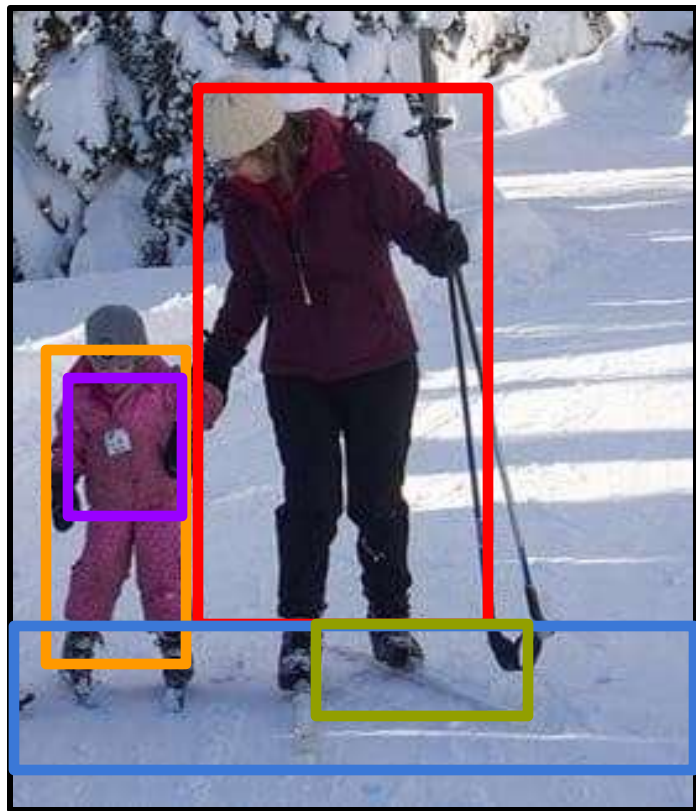


*A woman standing on a
beach holding a surfboard*

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<https://pixabay.com/en/teddy-plush-bears-cute-teddy-bear-1623436/>
<https://pixabay.com/en/surf-wave-silhouette-sport-art-1668716/>
<https://pixabay.com/en/woman-female-model-porrait-adult-983967/>
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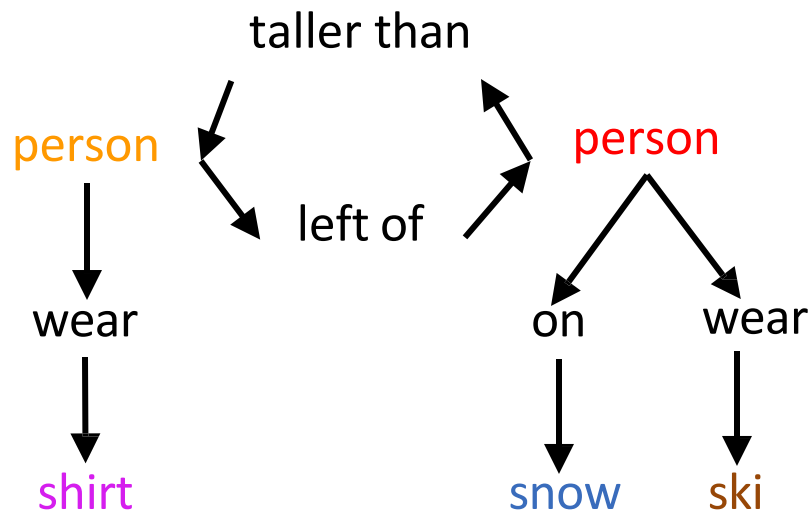
Captions generated by Justin Johnson using [NeuralTalk2](#)

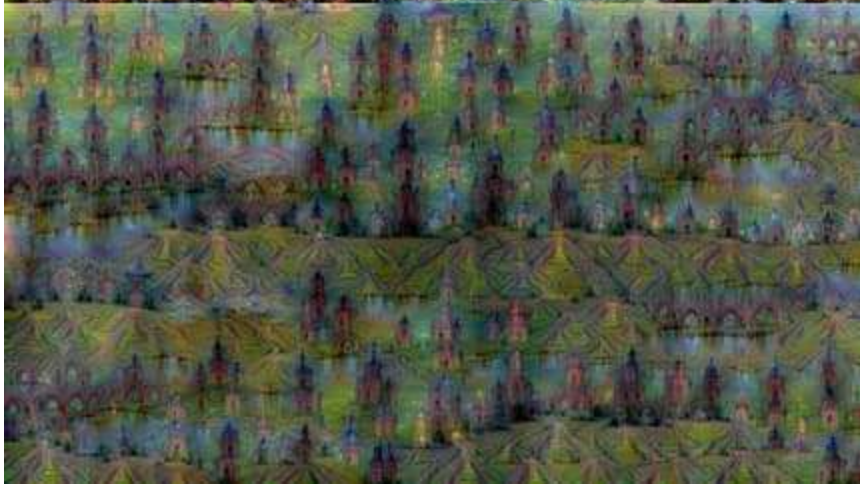
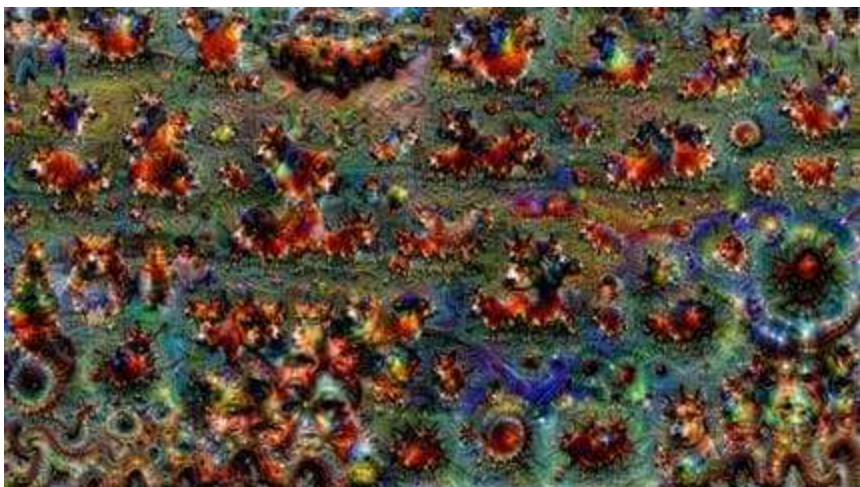
2012 to Present: Deep Learning is Everywhere



Results:

spatial, comparative, asymmetrical, verb,
prepositional





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Bokeh image is in the public domain
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Mordvinsev et al, 2015
 Gatys et al, 2016

2012 to Present: Deep Learning is Everywhere

TEXT PROMPT

an armchair in the shape of an avocado. an armchair imitating an avocado.

AI-GENERATED IMAGES



2012 to Present: Deep Learning is Everywhere

TEXT PROMPT

an armchair in the shape of a peach. an armchair imitating a peach.

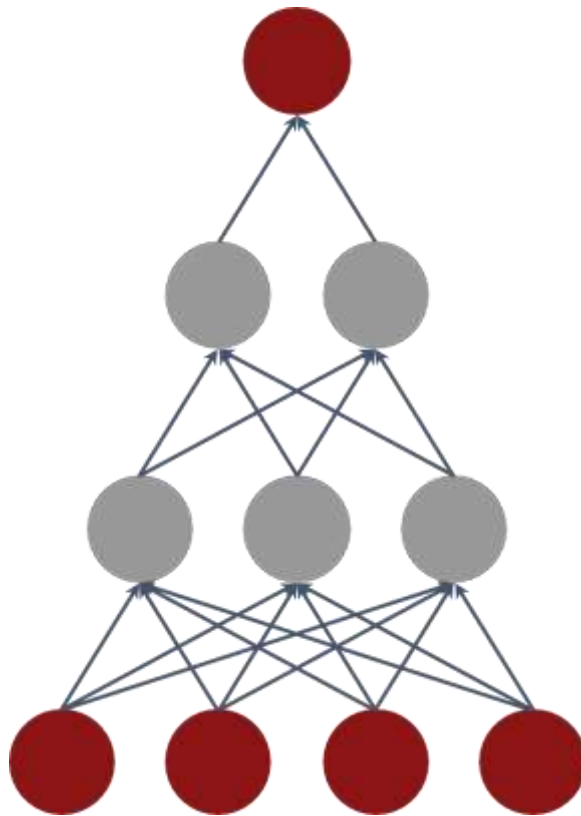
AI-GENERATED IMAGES





Computation

April 1, 2025



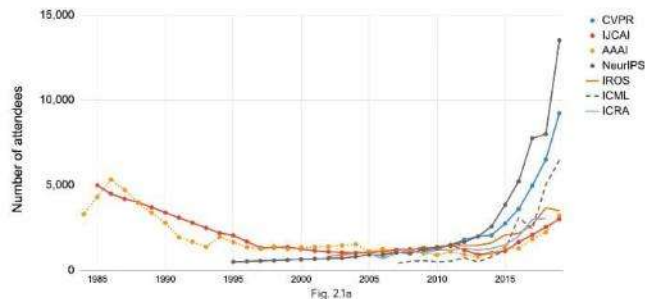
Algorithms



Data

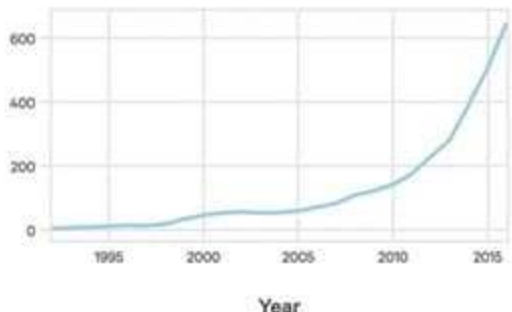
AI's Explosive Growth & Impact

Attendance at large conferences (1984-2019)
Source: Conference provided data



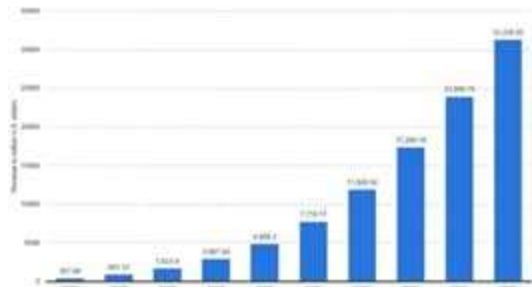
**Number of attendance
At AI conferences**

Source: The Gradient



**Startups Developing AI
Systems**

Source: Crunchbase, VentureSource, Sand
Hill Econometrics



**Enterprise Application AI
Revenue**

Source: Statista

Despite the successes, computer vision still has a long way to go

Computer Vision Can Cause Harm

Harmful Stereotypes

Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
Microsoft	94.0%	79.2%	100%	98.3%	20.8%
FACE++	99.3%	65.5%	99.2%	94.0%	33.8%
IBM	88.0%	65.3%	99.7%	92.9%	34.4%

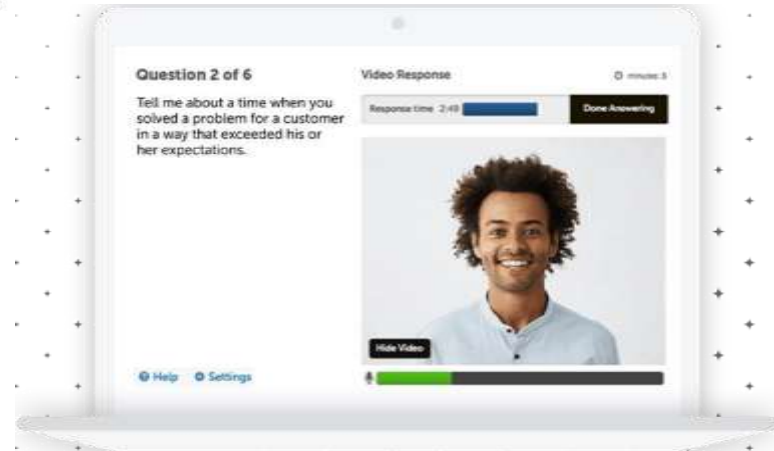


Affect people's lives

Technology

A face-scanning algorithm increasingly decides whether you deserve the job

HireVue claims it uses artificial intelligence to decide who's best for a job. Outside experts call it 'profoundly disturbing.'



Source: <https://www.washingtonpost.com/technology/2019/10/22/ai-hiring-face-scanning-algorithm-increasingly-decides-whether-you-deserve-job/>
<https://www.hirevue.com/platform/online-video-interviewing-software>

Example Credit: Timnit Gebru

Computer Vision Can Save Lives

**How to take care of seniors
while keeping them safe?**



Early Symptom Detection
of COVID-19



Monitor Patients with
Mild Symptoms



Manage Chronic Conditions



Versatile



Mobility



Infection



Sleep



Diet



Scalable



Low-cost



Burden-free



And there is a lot we don't know how to do



https://fedandfit.com/wp-content/uploads/2020/06/summer-activities-for-kids_optimized-scaled.jpeg



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Image Classification: A core task in Computer Vision

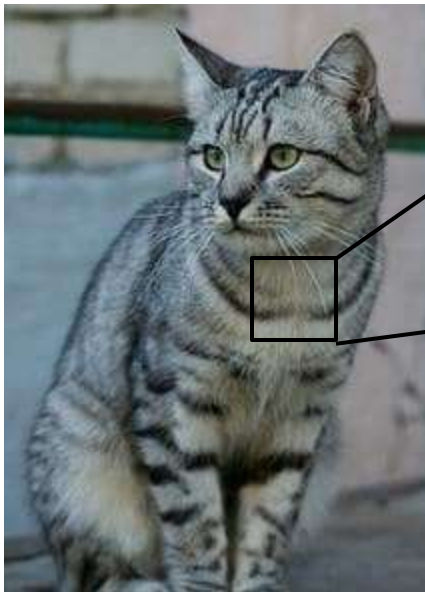


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(assume given a set of possible labels)
{dog, cat, truck, plane, ...}

—————→ cat

The Problem: Semantic Gap



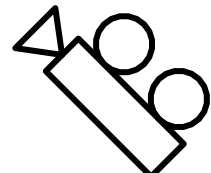
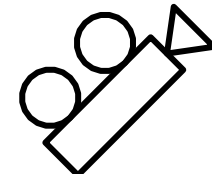
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What the computer sees

An image is a tensor of integers
between $[0, 255]$:

e.g. $800 \times 600 \times 3$
(3 channels RGB)



17490	152	280	133	84	90	90	183	121	139	184	97	93	271
17500	93	80	146	284	79	90	182	99	183	121	139	184	94
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17700	95	85	85	133	113	127	188	90	182	99	183	95	181
17800	94	84	94	84	84	84	84	84	84	84	84	84	84
17900	134	180	55	35	85	84	54	84	87	132	129	90	74
18000	137	137	147	185	35	85	93	52	74	84	142	93	85
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18400	137	137	147	185	35	85	93	52	74	84	142	93	85
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19400	137	137	147	185	35	85	93	52	74	84	142	93	85
19500	137	137	147	185	35	85	93	52	74	84	142	93	85
19600	137	137	147	185	35	85	93	52	74	84	142	93	85
19700	137	137	147	185	35	85	93	52	74	84	142	93	85
19800	137	137	147	185	35	85	93	52	74	84	142	93	85
19900	137	137	147	185	35	85	93	52	74	84	142	93	85
20000	137	137	147	185	35	85	93	52	74	84	142	93	85

Challenges: Illumination



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Challenges: Background Clutter



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Challenges: Occlusion



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Challenges: Deformation



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Challenges: Intraclass variation



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Challenges: Context



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Challenges: Context

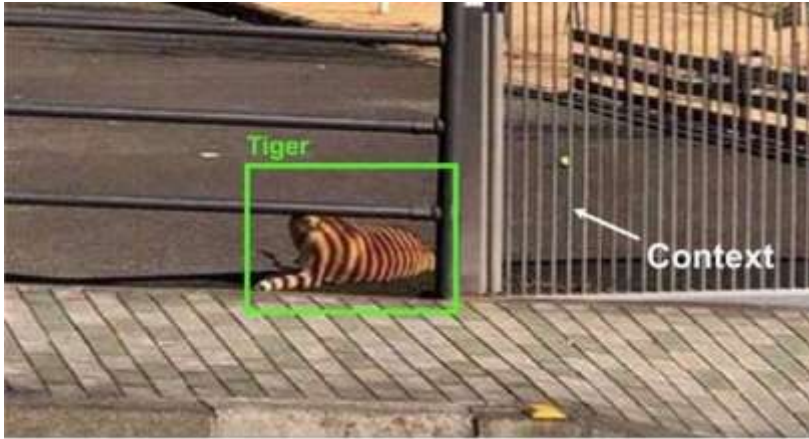


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Challenges: Context

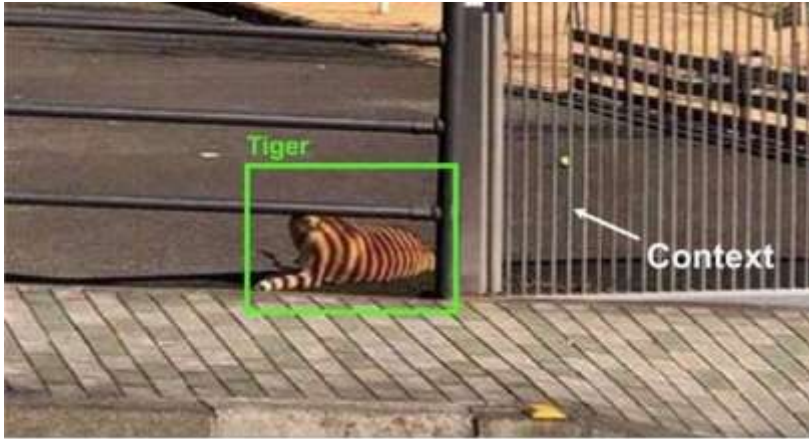


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Modern computer vision algorithms



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